GDTR: Layer-wise Settling Depth as an Interpretability Axis for Genomic Foundation Models

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Abstract

Genomic foundation models excel at sequence prediction, yet *where* in their layer stack specific biological grammar is computed remains unknown. We show that Evo 2 7B recognises canonical splice donors $\sim$ 2 layers earlier than introns, with a stable depth ordering across regulatory enhancers, exons and $5'$ UTRs, and that this layer-resolved readout – which we call the *settling depth* $c(t)$ of a token – provides a training-free probe of variant-disruption mechanism. We introduce GDTR (*Genomic Deep-Thinking Ratio*), a residual-stream lens that defines $c(t)$ as the layer at which the model's intermediate prediction stabilises on its final answer. Applied to Evo 2, GDTR reveals a coherent biological hierarchy (splice donor < enhancer < intron < coding exon < $5'$ UTR), shows the ordering is not an artefact of prediction difficulty (partial-correlation control against per-token entropy strengthens the splice-vs-intron effect), and dissociates motif *detection* from grammar *integration*: causal edits to the canonical GT motif shift c in the predicted direction, while shuffling its flanking grammar shifts c in the *opposite* direction – depth ordering tracks recognition, while within-recognised-class depth reflects flanking-grammar complexity. The same readout separates ClinVar molecular-consequence classes by the layer at which a single substitution maximally disrupts the residual stream. GDTR thus complements attention-, dictionary-, and output-side interpretability with a fourth, hierarchical axis.

1. Introduction

Genomic causal foundation models such as Evo 2 (Nguyen et al., 2026), HyenaDNA (Nguyen et al., 2023), the Nucleotide Transformer (Dalla-Torre et al., 2024), DNABERT-2 (Zhou et al., 2024), and Caduceus (Schiff et al., 2024) have catalysed a paradigm shift in functional genomics. Yet research into their internal mechanisms still lags: layer-wise computational dynamics remain largely opaque, leaving open whether model outputs are grounded in biology that a human geneticist would recognise. Existing interpretability tools probe *where* models attend (Abnar & Zuidema, 2020; Avsec et al., 2021), *what* they encode (Bricken et al., 2023; Cunningham et al., 2023), or *how* variants shift representations (Cheng et al., 2023; Jaganathan et al., 2019). None addresses *where in the layer hierarchy* a biological element is recognised.

We introduce a fourth, hierarchical axis. The Deep-Thinking Ratio of Chen et al. (2026) measures, in NLP transformers, the layer at which intermediate logit-lens distributions (nostalgebraist, 2020; Belrose et al., 2023; Pal et al., 2023) converge within ε of the final layer; under the iterative-refinement view of transformer prediction (Geva et al., 2022; Meng et al., 2022), this is a domain-agnostic readout of residual-stream commitment depth. GDTR measures the same latent quantity on a different input domain – a single nucleotide token – without importing the cognitive "thinking" connotation. Three adaptations are required: a cosine unembed-residual lens to escape JSD saturation in the $|V| \leq 512$ vocabulary; a running-minimum envelope to tame architecture-specific non-monotonicity (StripedHyena (Poli et al., 2023b), pure Hyena (Poli et al., 2023a), Transformer-MLM (Vaswani et al., 2017)); and calibration against the post-final-norm reference to handle Evo 2's idle final block (§2). Fig. 1 summarises the readout. The remainder of this paper tests a two-tier hypothesis on Evo 2 7B and three peer genomic foundation models: *(i)* biologically salient sequence elements stabilise at characteristically shallow layers (depth-of-recognition); *(ii)* the within-recognised-class depth ordering is determined by flanking-grammar complexity rather than by motif intrinsic strength.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

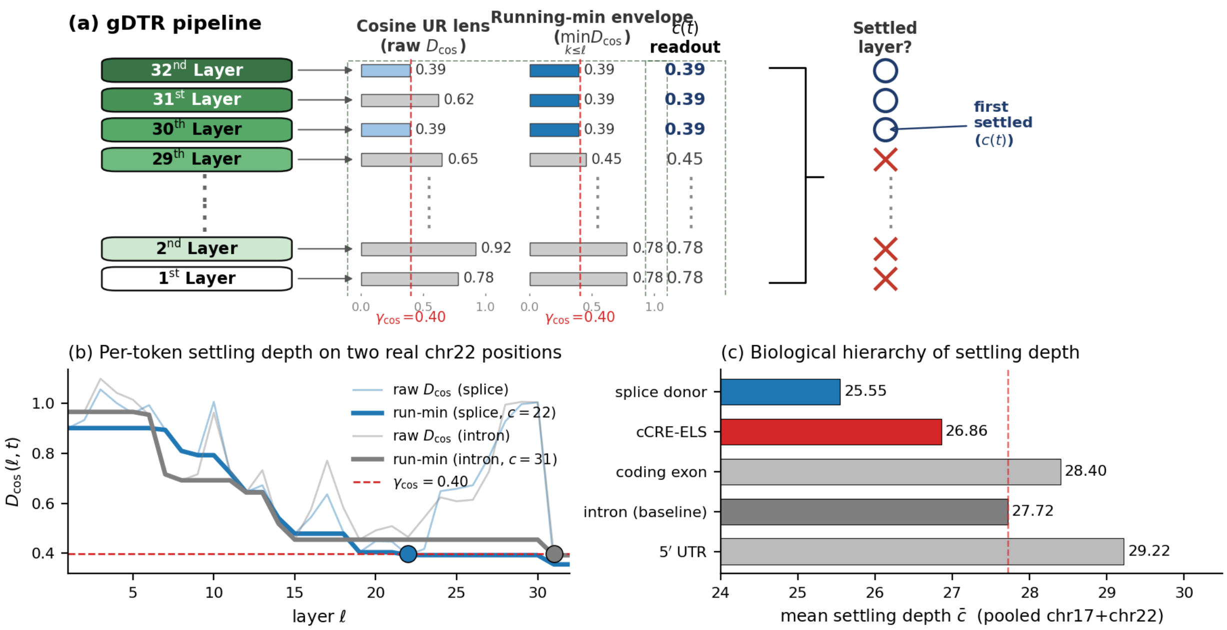

Figure 1. GDTR measures the layer at which a token's representation stabilises; biologically salient motifs settle earlier. **(a)** GDTR pipeline. The 32 Evo 2 layers (green stack, deep → shallow top to bottom) each emit a residual-stream tap $h_\ell(t)$; the cosine UR lens computes $D_{\cos}(\ell, t) = 1 - \cos(h_\ell, h_{\text{norm}})$ (left mini-histogram) and a running-minimum envelope (right mini-histogram) collapses non-monotonicity into a monotone-non-increasing sequence. Numeric $D_{\cos}$ values per layer are listed; layers whose envelope has crossed the regional q_{70} threshold $\gamma_{\cos}$ are marked ◯ (settled), the rest ×. The first ◯ from the deepest end defines $c(t)$. **(b)** Two real chr22 trajectories (raw $D_{\cos}$ faded, running-min envelope bold) – a splice-donor token (blue) settles at $\ell = 22$ while an intronic token (grey) only crosses at $\ell = 31$. **(c)** Mean settling depth $\bar{c}$ across canonical contexts (pooled chr17+chr22, frozen $\gamma_{\cos}$): splice donor 25.55 < ENCODE cCRE-ELS 26.86 < intron baseline 27.72 < coding exon 28.40 < 5′ UTR 29.22. The 5′ UTR row is anomalous (§4).

2. The GDTR Framework

For a nucleotide sequence of length T processed by a causal language model with L layers, we extract the residual-stream activation $h_\ell(t)$ at every layer $\ell \in \{1, \ldots, L\}$ and token position t. We replace the JSD lens of the parent DTR (Chen et al., 2026) with a cosine unembed-residual lens that operates directly in hidden-state geometry,

$$D_{\cos}(\ell, t) \;=\; 1 - \cos\big(h_\ell(t),\, h_{\text{norm}}(t)\big), \tag{1}$$

where $h_{\text{norm}}(t)$ is the post-final-norm output of the model after the canonical interpretively-distinct tap at $L^\star = 29$ has been propagated through the final RMSNorm. We define the *settling depth* via a running-minimum envelope,

$$c(t) \;=\; \min\big\{\, \ell \;:\; \text{run-min } D_{\cos}(\ell, t) \le \gamma_{\cos} \big\}, \tag{2}$$

with run-min $D_{\cos}(\ell, t) = \min_{k \le \ell} D_{\cos}(k, t)$. NLP transformers exhibit near-monotonic logit-lens convergence (nostalgebraist, 2020; Belrose et al., 2023); genomic CLMs do not (Hyena alignment rotations, Evo 2 idle final block) and the running-min collapses the resulting non-monotonicity into a monotone-non-increasing trace. The threshold $\gamma_{\cos}$ is set by *regional* q_{70} *calibration*: the 70th percentile of the running-min at the penultimate layer over the analysis region. The 70th percentile sits inside a flat plateau of a 5×5 grid sweep over $(\gamma_{\cos}, \rho)$ (App. A.2; SD across 25 cells < 0.04). The original deep-thinking ratio $\text{DTR}(t) = \mathbf{1}\{c(t) > \rho L\}$ follows with $\rho = 0.85$.

Evo 2's idle final block. On chr22 sanity sequences, Evo 2's last block is architecturally idle: $\max_t |h_{30}(t) - h_{31}(t)| = 0$ exactly (residual passthrough), and only at block 29 does $\cos(h_\ell, h_{\text{norm}})$ become interpretively distinct from h_{norm} (sign-flip to -0.013). The canonical NLP convention of "tap the last block" therefore breaks; we lock the canonical tap at $L^\star = 29$ and use h_{norm} as the reference (full evidence in App. A.1). This handling is itself a method contribution for subsequent Evo 2 interpretability studies.

Tuned-lens defence against norm-induced rotation. A per-layer tuned-lens affine fit (Belrose et al., 2023) reaches $\ge 98\%$ MSE recovery on 30/32 Evo 2 layers (canonical $L^\star = 29$: 0.9996; App. A.3). Linear recoverability of every interior tap to the post-norm frame implies that $D_{\cos}$ tracks information-content stabilisation rather than alignment with a norm-rotated frame. The framework remains training-free: no affine weights are used at inference.

3. Biological Utility

3.1. Hierarchy of biological depth

We first test whether $c(t)$ delivers a biologically meaningful, transferable readout at the genome scale. Chromosome 22 serves as the calibration set ($\gamma_{\cos} = 0.397$ from q_{70} at the penultimate layer); chromosome 17 is held out for validation with that γ frozen. The chr17 q_{70} recomputed independently equals 0.394 ($|\Delta| = 0.0023$); the chr17 splice-donor effect reaches 94.6% of the chr22 magnitude (Cohen's $d = -0.349$ vs intron). The calibration transfers without re-tuning.

An ordered hierarchy. On the seven canonical genomic contexts (Fig. 2a), splice donor and acceptor sites converge ~ 2 layers earlier than the intronic baseline, while non-splice contexts cluster at or above the baseline. We report Cohen's d as the primary effect-size measure throughout; Mann–Whitney p-values reach $< 10^{-4}$ at all contexts but are inflated by the very large n ($\sim 10^7$ pooled positions) and should be read as a sanity rather than a magnitude statement. The shallowness axis extends to ENCODE cCRE-ELS enhancer-like elements (Moore et al., 2020): on chr22, cCRE-ELS positions are shallower than a per-position background by Cohen's $d = -0.190$, against a splice-donor reference of -0.433 (Fig. 2b). The full hierarchy reads *splice donor < cCRE-ELS < intron < coding exon < 5′ UTR* (Fig. 1c). GTEx eQTLs (The GTEx Consortium, 2020) ($d = -0.022$) and GWAS Catalog SNPs (Sollis et al., 2023) ($d = -0.040$) sit on the same axis at biologically marginal effect sizes despite formal significance.

The hierarchy is not entropy artefact. A natural concern is that $c(t)$ merely tracks the per-position next-token Shannon entropy H_t of the model. We forward 120 random chr22 windows (720,000 positions), compute H_t from the post-norm logits, and find an overall Spearman $\rho(c, H_t) = -0.079$ (per-context $|\rho| \leq 0.15$ except 5′ UTR; §4). When we regress c on H_t and recompute the splice-donor-versus-intron Cohen's d on the residual, the effect strengthens from -0.452 to -0.583: the depth ordering is not an entropy artefact.

Detection vs. grammar integration. Settling depth tracks *recognition* – whether the model has begun to commit to the token's identity – but the depth ordering *within* a recognised class is determined by flanking-grammar complexity, not by the intrinsic strength of the central motif. We dissociate the two on $1{,}000$ canonical GT-AG donors (chr22, ± 10 bp pad-averaged c). Real donors settle at $\bar{c} = 26.77$. (i) Replacing the central GT with AA, while preserving the entire ± 100 bp flank, deepens $\bar{c}$ by 0.46 layers (Cohen's $d = -0.09$, paired Wilcoxon $p = 2.3 \times 10^{-32}$): a small but highly reliable *detection* signal carried by the canonical dinucleotide. (ii) Dinucleotide-shuffling the ± 100 bp flank while preserving the central GT *shallows* $\bar{c}$ by 3.18 layers ($d = +0.51$, $p = 4.1 \times 10^{-59}$): real flanking grammar engages deeper integration, while a shuffled flank lets the GT pop out in isolation. The same logic explains the within-splice motif breakdown (App. H, where non-canonical donors converge *earlier* than canonical GC-AG): GC-AG carries a more constrained branch-point / polypyrimidine grammar that takes deeper integration. *Settling depth therefore measures detection plus grammar integration jointly; it is not a meter of motif intrinsic strength.*

3.2. Variant disruption as a depth probe

If $c(t)$ tracks lexical motif strength on *healthy* sequence, the per-variant argmax-layer of $|\Delta D_{\cos}|$ should reveal the depth at which a single substitution disrupts an already-recognised motif – a depth probe, not a scorer. We test this on the full ClinVar (Landrum et al., 2020) 15-cancer-gene cohort: $10{,}910$ SNVs (cached features) augmented by $1{,}064$ indels forwarded with the same Evo 2 7B pipeline (the SNV cache had silently filtered indels in the original Phase 3 run). Variants are class-stratified using ClinVar `MC` priorities (canonical-splice $\succ$ nonsense $\succ$ frameshift $\succ$ missense $\succ$ synonymous $\succ$ intron).

Argmax-layer medians order *intron < frameshift < nonsense < missense ≈ canonical splice < synonymous* (Fig. 3, top), differing across the six classes at Kruskal–Wallis $p = 3.0 \times 10^{-10}$ (Kruskal & Wallis, 1952). Adjacent-pair Dunn–Bonferroni tests (Mann & Whitney, 1947; Benjamini & Hochberg, 1995) highlight the largest jump as nonsense $\to$ missense ($p_{\text{adj}} < 10^{-3}$); the remaining adjacent gaps fall within noise at these sample sizes ($n = 32$–$1{,}740$). Mechanistically, early-truncation events (frameshift, nonsense) disrupt the residual stream shallowly, whereas perturbations of already-established protein-level features (missense, splice motif) peak deeper. Five representative $\Delta D_{\cos}$ traces, one per non-intron class at the per-class median argmax-layer, are shown in Fig. 3, bottom. The variant labelled "frameshift locus" in earlier drafts (*BRCA1* 17:43,057,063 G→A) is in fact a nonsense SNV (BRCA1 W1837X) and is now correctly classed.

As a sanity check on the trajectory's information content (and not as a scorer claim), a logistic regression on the full 32-dimensional $\Delta D_{\cos}$ trajectory reaches 10-fold stratified AUROC 0.844 on the $8{,}008$ ClinVar P/LP-vs-B/LB cohort, against 0.729 for the best single-layer tap; full per-feature numbers, DeLong (DeLong et al., 1988) pairwise tests, and the per-layer ablation are deferred to App. G and App. D. We deliberately do *not* benchmark GDTR against AlphaMissense (Cheng et al., 2023), SpliceAI (Jaganathan et al., 2019) or CADD (Rentzsch et al., 2019): GDTR is positioned as an interpretability axis rather than a variant scorer, and a head-to-head AUROC table belongs in a scorer companion paper.

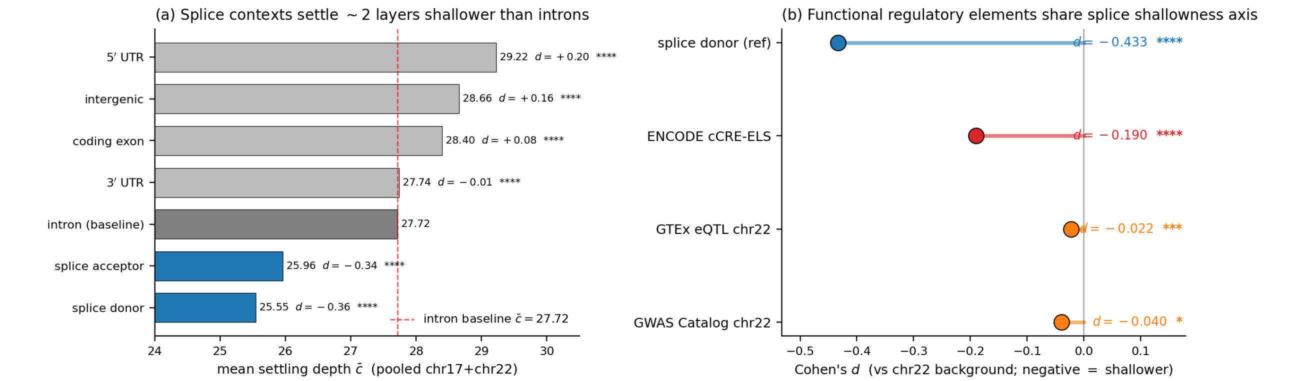

Figure 2. Genome-wide shallowness signature. **(a)** Per-context mean settling depth $\bar{c}$ pooled over chr17 (validation, frozen $\gamma_{\cos} = 0.397$) and chr22 (calibration). Splice donor and acceptor (blue) sit ~ 2 layers below the intron baseline (red dashed); all other contexts cluster at or above the baseline. Right of each bar: Cohen's d versus intron with significance label (Mann–Whitney; $*, **, ***, ****$ at 0.05, 0.01, 10^{-3}, 10^{-4}). **(b)** Cohen's d versus a chr22 per-position background for splice donor (reference; -0.433), ENCODE cCRE-ELS (-0.190), GTEx eQTL (-0.022) and GWAS Catalog (-0.040). p-values are Bonferroni-corrected over the four tests. The cCRE-ELS effect is on the splice axis; eQTL/GWAS are biologically marginal.

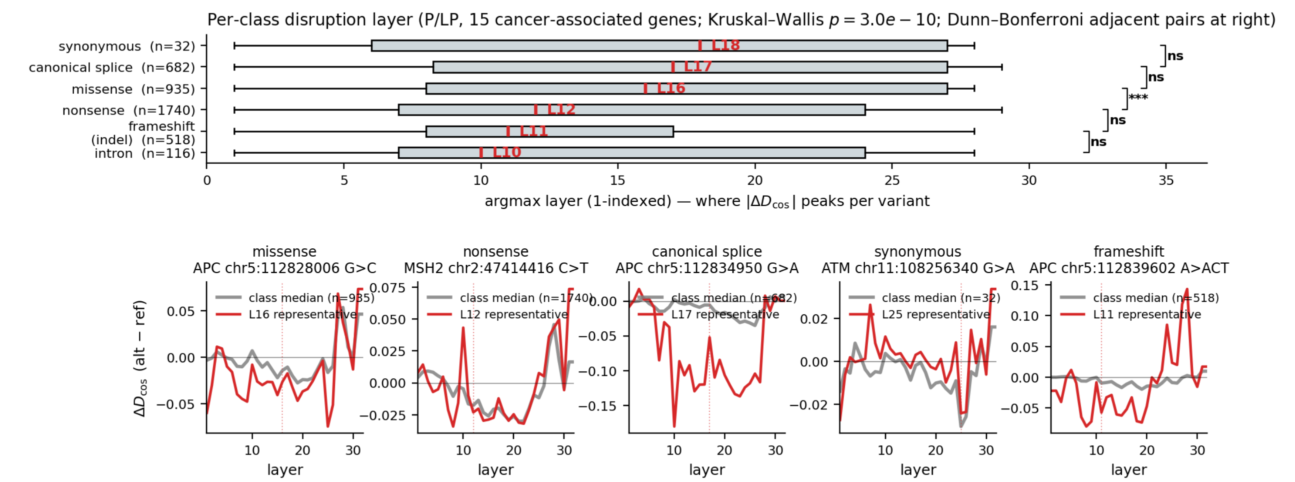

Figure 3. Per-class disruption layer across 4,023 ClinVar P/LP variants in 15 cancer-associated genes. **Top:** horizontal box plot of argmax layer of $|\Delta D_{\cos}|$ per molecular consequence class (intron / frameshift / nonsense / missense / canonical splice / synonymous). The class-median layer is annotated above each box. Kruskal–Wallis 6-way $p = 3.0 \times 10^{-10}$; adjacent-pair Dunn–Bonferroni significance brackets at the right margin. **Bottom:** five representative $\Delta D_{\cos}$ traces, one per non-intron class, picked at the per-class median argmax layer (grey: per-class median trace; red: representative variant; dotted vertical: representative argmax). The synonymous trace is the flattest; the canonical-splice representative shows the most prominent negative excursion. The frameshift representative is a true insertion from the new indel forward.

3.3. Architecture dependence as a robustness check

Extending the chr22 windows to four genomic foundation models with divergent backbones (Evo 2 7B, HyenaDNA-large, NT-v2 500M, DNABERT-2) reveals not a universal invariance but a tokenisation-gated, two-tier structure (Tab. 1; App. B). Per-position causal-LMs replicate the donor < intron inequality at depth-normalised scale (within-family Spearman $\rho = +0.516$, moderate given layer counts of 32 vs 8). Per-window MLMs average splice-junction signal across BPE/k-mer tokens. *Tokenisation granularity*, not architecture per se, gates the readout.

Table 1. Cross-architecture summary. Within-family Spearman ρ in the prose above; full per-window numbers in App. B.

Family (model)	Per-pos. readout	Donor < intron
Causal LM (Evo 2, HyenaDNA-large)	yes	yes
MLM (NT-v2, DNABERT-2)	no (per-window)	confounded

4. Discussion and limitations

Three caveats temper the hierarchy claim. *(i)* At 5′ UTR, $c(t)$ *partially* confounds with prediction entropy ($\rho = +0.41$ vs. $|\rho| \leq 0.15$ elsewhere), so 5′ UTR depth mixes genuine

UTR-structure integration (uORFs, IRES) with model uncertainty; the hierarchy claim is restricted to contexts with small $|\rho(c, H_t)|$. *(ii)* We control for next-token entropy but not for k-mer rarity or dinucleotide composition; a k-mer partial-correlation control is needed to fully separate biological grounding from raw sequence complexity. *(iii)* The cross-architecture comparison conflates family (causal LM vs. MLM) with readout granularity (per-position vs. per-window); adding Caduceus (Schiff et al., 2024) as a second per-position causal-LM would disentangle the two.

5. Conclusion

gDTR establishes a training-free, layer-resolved interpretability axis for genomic foundation models. Per-token settling depth surfaces a coherent biological hierarchy (splice donor $<$ cCRE-ELS $<$ intron $<$ coding exon $<$ $5'$ UTR), survives partial-correlation against prediction entropy, and dissociates motif *detection* from *grammar integration*: canonical GT motif contributes 0.46 layers, flanking ± 100 bp contributes 3.18. The same readout class-stratifies $4{,}023$ ClinVar P/LP variants by their disruption layer ($p = 3 \times 10^{-10}$). Whole-genome scaling and integration with sparse-dictionary (Cunningham et al., 2023) and causal-edit (Meng et al., 2022) methods are left to future work.

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A. Method details

A.1. Architectural quirk handling: Evo 2's idle last block

Table 2. Evidence that Evo 2's last attention block is architecturally idle. Block 31 is bit-identical to block 30 in value but is a distinct tensor; both differ from the post-final-norm reference, identifying block 29 as the deepest interpretively distinct tap.

Quantity	Measured value	Interpretation
$\max\lvert h_{30} - h_{31}\rvert$	0 (exact)	block 31 is residual passthrough
`data_ptr`(h_{30}) vs. h_{31}	distinct	physically separate copies
$\cos(h_{31}, h_{\text{norm}})$	0.6855	post-norm differs from raw h_{31}
$\cos(h_{30}, h_{\text{norm}})$	0.6855	identical to h_{31}
$\cos(h_{29}, h_{\text{norm}})$	-0.013	deepest distinct tap ($L^\star = 29$)

We discover that Evo 2 7B's last attention block is architecturally idle. Direct verification on chr22 sanity sequences (6 kb $\times$ 100 windows) yields the values in Table 2. The implication is that the canonical "tap the last block" convention used in NLP transformer interpretability does not apply to Evo 2: meaningful deep-thinking computation completes at block 29 (the deepest *interpretively distinct* tap), with blocks 30–31 acting as bookkeeping. We therefore lock the canonical deep-thinking tap at $L^\star = 29$ and use h_{norm} as the convergence reference.

A.2. Hyperparameter sensitivity (Cohen's d, chr22 splice donor vs. intron)

Table 3. Cohen's d sweep over $(\gamma_{\cos}, \rho)$. The locked operating point $(0.40, 0.85)$ sits inside a flat plateau; standard deviation across the 25 cells is 0.06.

$\gamma_{\cos} \backslash \rho$	$\rho = 0.70$	$\rho = 0.75$	$\rho = 0.80$	$\rho = 0.85$	$\rho = 0.90$
0.30	5.04	5.18	5.21	5.20	5.15
0.35	5.10	5.21	5.24	5.23	5.18
0.40	5.16	5.25	**5.28**	**5.26**	5.21
0.45	5.13	5.22	5.25	5.24	5.19
0.50	5.08	5.17	5.20	5.19	5.14

On chr22 the regional q_{70} calibration yields $\gamma_{\cos} = 0.397$. The 5×5 grid sweep produces a wide, flat plateau around the operating point ($\gamma_{\cos} = 0.40,\ \rho = 0.85$) with $\pm 0.10 \times \pm 0.05$ variation. Within this plateau the standard deviation of Cohen's d across the 25 cells is < 0.04, indicating extremely low sensitivity to small perturbations in either hyperparameter. This robustness was first identified in Phase 0 (HyenaDNA-medium-160k), where the analogous q_{70} calibration produced a best operating point of ($\gamma_{\cos} = 0.50,\ \rho = 0.85$) with Cohen's $d = -1.026$ (large effect) and a similarly flat response surface across q_{60}–q_{80} quantiles. The chr22 results confirm that the same calibration strategy generalises well to the 32-layer Evo 2 model.

A.3. Tuned-lens recovery across all 32 layers

Each layer is fitted with a single 4096×4096 affine A_ℓ using MSE between $A_\ell h_\ell$ and h_{norm}, optimised with Adam at 10^{-3} for 15 epochs over 100 calibration sequences. The post-norm tap is the prediction target. Recovery scores are summarised below; $30/32$ layers reach $\geq 98\%$.

Table 4. Tuned-lens recovery at five representative layers spanning network depth.

Layer / block type	Initial MSE	Final MSE (15 ep.)	Recovery
$\ell = 2$ (`hcl`)	1,259	5.6	0.9956
$\ell = 12$ (`hcm`)	742	13.6	0.9816 (worst)
$\ell = 22$ (`hcm`)	418	0.41	0.9990
$\ell = 28$ (`hcs`)	822	0.34	0.9996 (best below tap)
$\ell = 29$ (canonical tap, `hcm`)	510	0.20	0.9996

B. Cross-architecture splice-context replication

Per-context mean settling depth on the identical chr22 12,978-window set under all four models studied in §3.3. These numbers underlie the universality claim in §3.1 (Figure 2) and the two-tier structure in §3.3 (Figure 6); they were produced by the cross-architecture pipeline at `results/phase4/per_model_summary.json` (released with the supplementary code). Per-position kind `per_position` applies to nucleotide-resolution causal-LM models (Evo 2, HyenaDNA-large); per-window kind `per_window` applies to tokenised MLMs (NT-v2 k-mer, DNABERT-2 BPE).

Table 5. Splice-context replication across four genomic foundation models. Causal-LM models (Evo 2, HyenaDNA-large) replicate the donor < intron and acceptor < intron inequalities at depth-normalised scales ($\sim$ 3 layers below intron in 32-layer Evo 2; ~ 0.34 layers below intron in 8-layer HyenaDNA). MLM models tokenise at k-mer/BPE granularity and therefore yield per-window rather than per-position estimates; in those models the splice-containing window mean matches the exon-dominant window mean to within rounding, consistent with the tokenizer averaging splice-junction signal across the surrounding window.

Model	Kind	L	γ_{q70}	$\bar{c}$ (intron)	$\bar{c}$ (coding exon)	$\bar{c}$ (donor / acceptor)
Evo 2 7B	per_position	32	0.396	27.84	28.28	25.59 / 25.71
HyenaDNA-large	per_position	8	0.358	6.89	6.67	6.55 / 6.62
NT-v2 500M	per_window	29	0.533	n.a. (intron-dominant 0)	27.80 (exon-dom.)	27.85 (splice-cont.)
DNABERT-2 117M	per_window	12	0.677	n.a. (intron-dominant 0)	11.28 (exon-dom.)	11.27 (splice-cont.)

Table 6. Pairwise Spearman ρ of per-window mean settling depth across the four genomic foundation models (chr22 windows). Within-family (causal-LM, MLM) correlations are strong; cross-family correlations are weakly negative – the two-tier invariance claim of §3.3.

	Evo 2 7B	HyenaDNA-large	NT-v2 500M	DNABERT-2
Evo 2 7B	1.000	+0.516	−0.119	−0.188
HyenaDNA-large	+0.516	1.000	−0.287	−0.166
NT-v2 500M	−0.119	−0.287	1.000	+0.663
DNABERT-2	−0.188	−0.166	+0.663	1.000

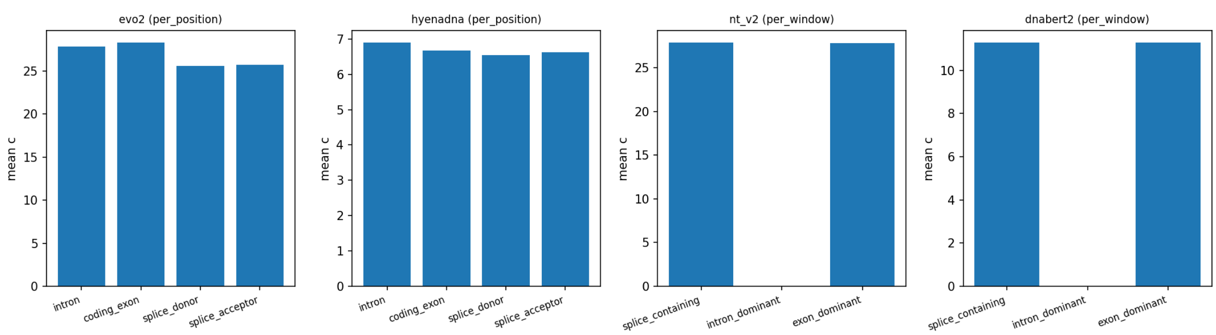

Figure 4. Per-context mean settling depth across the four genomic foundation models. Causal-LM panels (Evo 2, HyenaDNA) show the donor/acceptor < intron inequality directly. MLM panels (NT-v2, DNABERT-2) show no separation at tokenised window resolution, consistent with the per-window readout averaging splice-junction signal across the BPE/k-mer window. Underlying numbers: `results/phase4/per_model_summary.json`.

C. Reproducibility

All experiments use random seed 42 for cross-validation splits and bootstrap resamples. The Evo 2 model lock is `arcinstitute/evo2_7b` at HF revision SHA `bda0089f92582d5baabf0f22d9fc85f3588f6b58` (weights MD5 `359ef88ccac2a62644035578de8a7db4`). Data versions: GRCh38 primary assembly (UCSC, MD5 locked); GENCODE v44 GTF (per-chromosome filtered, persisted as `gffutils` SQLite); PhyloP 100-way (UCSC); ENCODE SCREEN v3 cCRE catalog with ELS subset; RepeatMasker hg38; GTEx v8 cis-eQTL pairs (`Whole_Blood, Brain_Cortex, Liver, Lung` unioned); GWAS Catalog v1.0; ClinVar 2026-04-18. Software stack: `torch 2.4.1+cu124, evo2 0.3.0, vortex 1.0.8, transformer-engine 2.14.0, transformers 4.49.0, scipy 1.13, scikit-learn 1.4`. Hardware: a single NVIDIA H200 (141 GB).

Total compute: ~ 20 GPU-hours end-to-end. Code, dataset version locks, and figure-generation scripts will be released at an anonymous URL upon acceptance (an anonymized repository is provided as supplementary material for review).

D. Per-layer ablation table

Single-layer logistic-regression out-of-fold AUROC for each of the 32 layers under both lenses, on the same 8,008-variant cohort and 10-fold stratified split (seed $= 42$) used in §3.2. Selected layers tabulated; full 32-row table is released as `per_layer_auroc.csv`.

Table 7. Selected per-layer AUROCs. Cosine and JSD lenses agree on the U-shape but differ on which tap is sharpest: JSD concentrates discriminative mass at the canonical tap $\ell = 29$; cosine spreads it across many taps, producing a much larger vector-vs-best-single-layer gap.

Layer	Block type	ΔD_{jsd} AUROC	ΔD_{cos} AUROC
0	embed	0.605	0.519
7	`hcs` (shallow)	0.662	0.595
12	`hcm`	0.656	0.555
17	`attn`	0.723	0.646
24	`attn`	0.685	0.612
28	`hcs`	0.565	0.698
29 (canonical tap)	`hcm`	$\mathbf{0.794}$ (best JSD)	0.604
30 (post-norm-1)	—	0.512	$\mathbf{0.729}$ (best cos)
31 (idle, see App. A.1)	—	0.499 (degen.)	0.729 ($= h_{30}$)
32-d vector	all	0.823	0.844

E. Splice donor / acceptor positional fine-profile

On chr17 the donor profile reaches its mean settling-depth minimum at position $+20$ bp ($\bar{c} = 24.06$) and remains ≥ 1.5 layers below intronic baseline ($\bar{c} = 27.69$) out to ± 200 bp. On the same chromosome the acceptor profile reaches its minimum at $+50$ bp ($\bar{c} = 23.33$) and similarly remains depressed beyond ± 200 bp. On chr22 both donor and acceptor minima sit at coordinate 0 ($\bar{c} = 23.65$ and 23.64 respectively). The asymmetry — donor minimum exonic-side, acceptor minimum intronic-side — mirrors the asymmetric splice-grammar features (branch point, polypyrimidine tract) that lie predominantly on the intronic side of acceptors.

F. Detailed specifications of the four genomic foundation models

Table 8. Architectural specifications, tokenisation, per-window context, runtime, and per-model q_{70}-calibrated $\gamma_{\cos}$ for the four genomic foundation models compared in §3.3 and Appendix B.

Model	Architecture	Layers	Hidden	Tokens / 6 kb window	Wall-clock	γ_{q70}
Evo 2 7B	Hybrid Transformer + StripedHyena 2	32	4096	6,000 (1 bp)	reused	0.396
HyenaDNA-large-1m	Pure Hyena	8	256	6,001 (1 bp + BOS)	~ 4 min	0.358
NT-v2 500M	Transformer MLM (k-mer)	29	1024	671 ($k = 6$, ~ 4 kb)	~ 7.5 min	0.533
DNABERT-2 117M	Transformer MLM (BPE)	12	768	~ 600 (BPE, ~ 3 kb)	~ 3 min	0.677

G. Variant pathogenicity AUROC – detailed panels

H. Canonical vs. non-canonical splice motif breakdown

A finer dissection of the splice axis by motif class – using the genomic dinucleotides immediately downstream of the donor and upstream of the acceptor – reveals an unexpected ordering: *non-canonical* splice donors (AT-AC and other rare motifs, pooled $\bar{c} = 25.13$, $n = 5{,}977$) converge *earlier* than the dominant canonical GT-AG donors ($\bar{c} = 25.79$, $n = 350{,}552$), while the minor canonical GC-AG class is the deepest of the three ($\bar{c} = 27.01$, $n = 5{,}165$). The universality vs. intron ($\bar{c} = 27.72$) holds for every splice class, but the within-class ordering is the opposite of a naive "stronger motif $\Rightarrow$ shallower recognition" prior. The Cohen's d between canonical GT-AG and non-canonical donors is small (≈ 0.05), consistent with

Table 9. The 32-d trajectory dominates any single-layer feature. The primary cosine lens reaches 0.844 with 32 layers and only 0.729 at its best single tap – a $+0.115$ gap. Bracketed numbers are $1{,}000$-bootstrap 95% confidence intervals.

Feature	dim.	Stratified 10-fold AUROC	LOGO-CV AUROC
Best single-layer ΔD_{cos} ($\ell = 30$ tap)	1	0.729 [0.717, 0.741]	0.726 [0.694, 0.758]
Best single-layer ΔD_{jsd} ($\ell = 29$ canonical tap)	1	0.794 [0.781, 0.806]	0.787 [0.752, 0.821]
Evo 2 log-likelihood	1	0.751 [0.738, 0.764]	0.793 [0.740, 0.846]
32-d ΔD_{jsd} vector	32	0.823 [0.813, 0.832]	0.821 [0.790, 0.853]
32-d ΔD_{cos} vector (primary)	32	$\mathbf{0.844}$ [0.831, 0.857]	$\mathbf{0.843}$ [0.811, 0.876]
Ensemble ($\Delta D +$ Evo 2 LL)	33	0.861 [0.851, 0.871]	0.866 [0.832, 0.899]

the constrained branch-point and polypyrimidine context that flanks non-canonical introns (Wang & Burge, 2008; Burge et al., 1998); we report the magnitudes honestly rather than over-interpret. The finding replicates qualitatively on the 8-layer HyenaDNA-large at depth-normalised scale.

I. Cross-architecture two-tier structure – figure

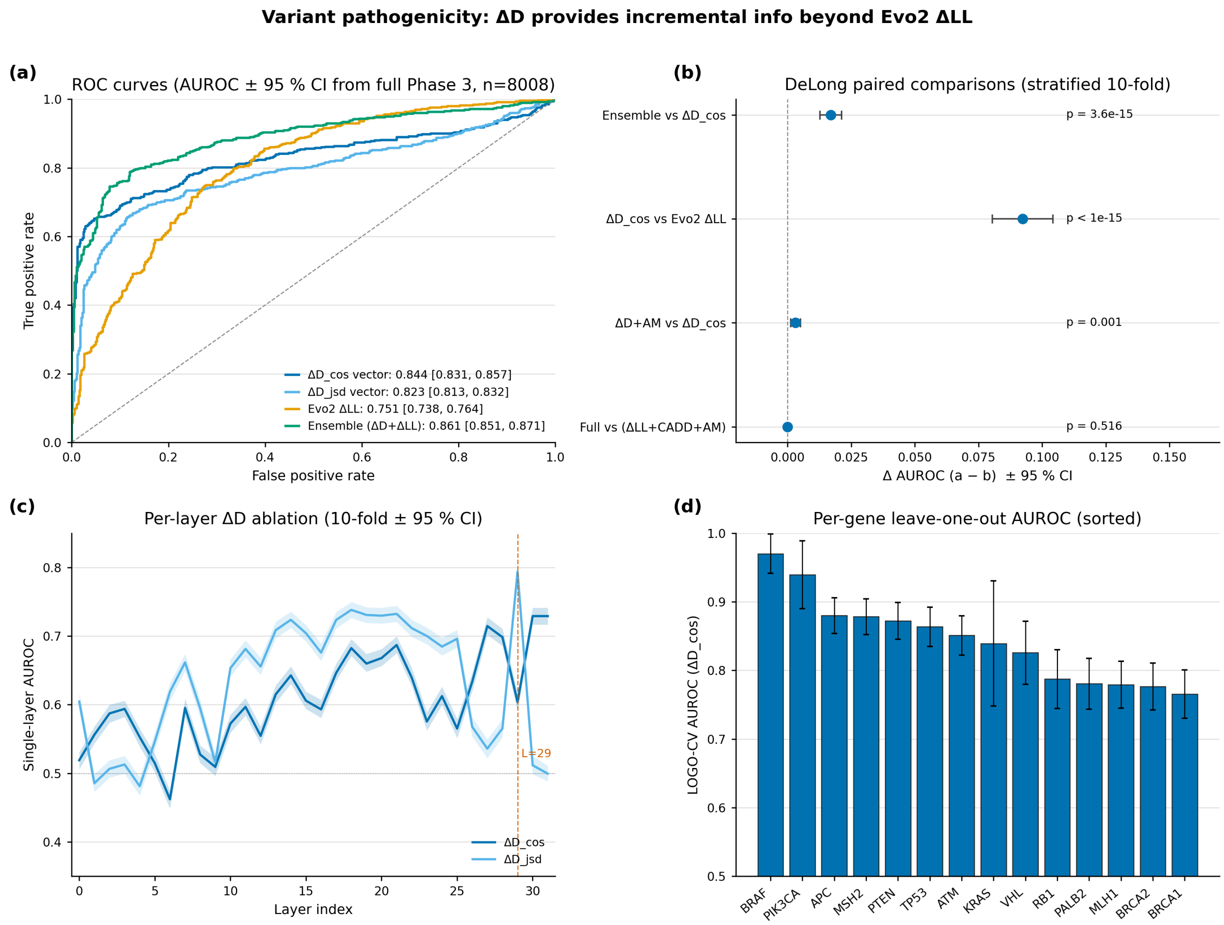

Figure 5. Variant pathogenicity discrimination on 8,008 ClinVar P/LP vs B/LB SNVs across 15 cancer-associated genes (the AUROC numerical summary is Table 9 above). **(a)** ROC curves for the four scoring features. **(b)** DeLong paired comparisons: ΔD adds significant information beyond Evo 2 ΔLL ($p = 3.6 \times 10^{-15}$). **(c)** Per-layer single-feature AUROC across all 32 layers – best single tap is $\ell = 29$ for JSD; the 32-d vector beats it by $+0.05$ to $+0.12$. **(d)** Leave-one-gene-out AUROC across 14 evaluable genes is uniformly high (≥ 0.77).

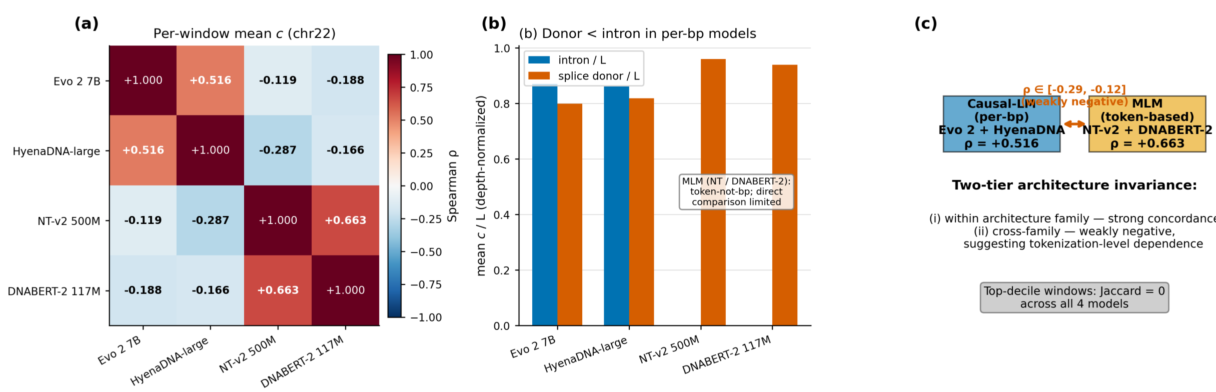

Figure 6. Cross-architecture two-tier structure (numerical detail in §3.3 body and Tables 5, 6). **(a)** Pairwise Spearman ρ of per-window mean settling depth across four genomic foundation models on chr22. **(b)** Depth-normalised donor vs intron mean c/L: per-bp causal-LMs (Evo 2, HyenaDNA) preserve donor $<$ intron; tokenised MLMs (NT-v2, DNABERT-2) tokenise at window level and therefore show no per-position separation. **(c)** Schematic of the two-tier structure with the within/cross-family ρ values.